EfficientNet-DenseNet Network with Bidirectional Cross-Attention and early GradCAM mechanism (ED-XAGN) for Brain Tumor Diagnosis in MRIs

Abhay Tyagi, Abhijeet, Varun Soni and Divyansh Ahuja

[a]*Dept. of Computer Science, Netaji Subhas University of Technology,*

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ABSTRACT

To enhance the accuracy of brain tumor diagnosis and treatment, reliance on MRI images is crucial. However, human error in manual diagnosis remains a concern, underscoring the urgency for robust TL (Transfer Learning) and CNN (Convolutional Neural Network) based techniques to enhance accuracy in discerning normal and malignant brain tumors. Our paper introduces a novel method, the EfficientNet-DenseNet Network with Bidirectional Cross-Attention and early GradCAM mechanism (ED-XAGN) model, aimed at improving classification accuracy. By integrating features from two complementary pre-trained CNN models: EfficientNetB0 and DenseNet-121, we have developed an architecture superior to comparable ones on the market. Our approach addresses challenges commonly found in tumor detection systems, including high processing demands, difficulty in discerning intricate details in images, and diminished efficacy due to noise or irrelevant data. Our architecture employs Gamma Correction techniques to reduce degradation in long-form feature capture, the twin-CNN backbone is augmented with a Bidirectional Cross-Attention module which allows both feature streams to query each other, it also introduces a novel fusion architecture called Adaptive Spatial-Channel Attention (ASCA) to optimize the weightages of local and far-form spatial features. We make use of a publicly available brain tumor dataset consisting of 7,200 MRI images, on which we apply various augmentations and standardization techniques to represent real-world conditions and enhance model compatibility.

1. Introduction

A tumor is an abnormal growth of cells that can be benign or malignant. When this growth occurs in the brain, it is known as a brain tumor (1). Brain tumors represent one of the most aggressive and clinically challenging neurological disorders, with their early and precise diagnosis being paramount for determining appropriate treatment strategies and improving patient outcomes. As the primary diagnostic modality for brain tumor detection, Magnetic Resonance Imaging (MRI) (2) offers exceptional soft tissue contrast and non-invasive visualization capabilities that are indispensable in clinical neurology. However, the manual interpretation of MRI scans remains fraught with significant challenges that compromise diagnostic reliability, including substantial inter-observer variability among radiologists, the inherent subjectivity of visual assessment, and the growing problem of diagnostic fatigue in high-volume clinical settings. These limitations become particularly pronounced when attempting to differentiate between common tumor types such as (3) (the most aggressive primary brain tumors), meningiomas (typically benign but potentially problematic tumors), and pituitary tumors (which present unique diagnostic challenges due to their anatomical location). The growing complexity of neuro-oncological diagnosis has spurred remarkable advances in artificial intelligence, particularly through the application of deep learning (4) for automated brain tumor classification. These computational approaches promise to overcome many of the limitations associated with human interpretation by providing consistent, quantitative, and rapid analysis of medical imaging data. However, despite their considerable potential, current deep learning-based classification systems face three fundamental challenges that limit their clinical applicability and diagnostic performance:

- *Sensitivity to Irrelevant Spatial Artifacts*: Standard CNN backbones process background cranial bone, soft tissue, and acquisition noise uniformly alongside parenchymal pathology, degrading feature representation purity.
- *Receptive Field and Contextual Bottlenecks*: Standard localized convolutional kernels struggle to capture broad multi-scale spatial dependencies and context across complex, heterogeneous tumor microenvironments.
- *Loss of Spatial Coordinate Geometry in Feature Fusion*: Naive feature fusion methods collapse 2D spatial feature maps via premature Global Average Pooling (GAP) prior to vector concatenation or element-wise addition, destroying 2D coordinate layouts and restricting explainability.
- *Predictive Opacity and Lack of Reliability Quantification*: Conventional point estimate deep classifiers do not quantify diagnostic uncertainty, offering clinicians

abhaytyagi2027@gmail.com (A. Tyagi); jitukhatkarabhijeet123@gmail.com (Abhijeet); varunsonigdgps@gmail.com (V. Soni); ppds@gmail.com (D. Ahuja)
ORCID(s):

no indication of when a model prediction is ambiguous or unconfident.

2. Related Work

Automated computer-aided diagnosis systems for brain tumor classification from MRI scans have evolved across several deep learning paradigms. Convolutional Neural Networks utilizing transfer learning (5) (6) remain foundational in neuro-imaging due to their ability to extract hierarchical spatial primitives from pre-trained natural image domains. Early studies evaluated standard architectures including AlexNet, VGG-16, and fine-tuned ResNet variants (5), demonstrating automated tumor detection on benchmark datasets like BraTS, but observing performance drops along complex infiltrative tumor margins. Standalone models such as DenseNet-121, DenseNet-201, InceptionV3, and Xception (10) have been benchmarked extensively on datasets like BR35H. However, individual CNN backbones often struggle to balance fine-grained local boundary extraction with broad multi-scale semantic context, driving researchers toward multi-stream feature fusion frameworks.

To enhance feature selectivity and suppress non-brain cranial noise, recent architectures incorporate attention mechanisms and Vision Transformers (ViTs) (7) (8) . Attention-Guided CNNs (AG-CNN) highlight pathological regions while suppressing healthy tissue background. Hybrid Transformer Enhanced CNNs (TECNN) (11) utilize self-attention blocks from Vision Transformers to model long-range spatial dependencies across axial slices. Similarly, ConvAttenMixer (9) combines convolutional mixer blocks with external self-attention mechanisms to differentiate glioma subtypes. While transformer based attention enhances spatial focus, applying pure Vision Transformers directly to raw pixel grids imposes heavy parameter burdens and high memory consumption.

Combining complementary representations from multiple deep backbones via feature fusion has emerged as an effective strategy to boost classification performance. Ensembles combining lightweight CNNs with post-fusion attention guidance and sparse autoencoder networks like TumorAwareNet (7) have demonstrated improvements over single-stream baselines.

2.1. Baseline Study

Most recently, Hekmat et al. (12) proposed the Attention-Fused MobileNet-LSTM (AFML) framework, combining a ViT-L/16 attention pre-filter with gamma contrast correction, parallel MobileNetV1 and MobileNetV2 feature extraction, post-GAP vector concatenation, and an LSTM recurrent unit for classification.

2.1.1. Architecture

1. *Transformer Attention Preprocessing*: Raw MRI frames are passed through a Vision Transformer (ViTL16) self-attention encoder to suppress non-brain anatomical structures and background noise.
2. *Photometric Luminance (Gamma) Calibration*: The attention-filtered representation undergoes gamma correction to calibrate contrast across dynamic ranges.
3. *Dual Backbone Extraction*: The contrast-calibrated image is fed simultaneously into two parallel pre-trained backbones: MobileNetV1 (16) (86 layers, 3.23 million parameters) and MobileNetV2 (17) (154 layers, 2.26 million parameters).
4. *Vector Flattening and Concatenation*: Output feature maps are flattened via Global Average Pooling (GAP) and merged in a 1D vector concatenation layer, yielding a 2,304-dimensional vector.
5. *Recurrent Sequence Modeling*: The concatenated vector is routed into a Long Short-Term Memory (LSTM) (18) recurrent network to capture inter-feature sequential dependencies prior to SoftMax probability projection.

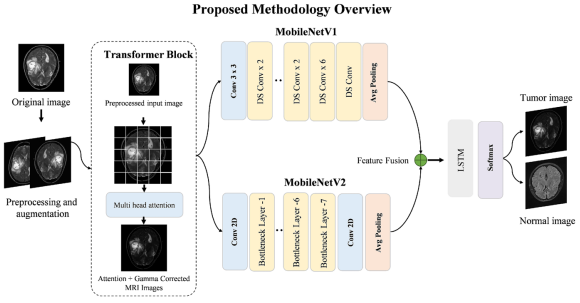


Figure 1: *AFML Architecture*

Although AFML achieved strong accuracy on binary benchmarks, its performance relied on concatenating structurally redundant lightweight encoders and unrolling static spatial vectors through an opaque LSTM module. Furthermore, on severely class-imbalanced datasets (such as the INbreast benchmark), AFML exhibited a major failure mode where malignant recall dropped to 0.12 (12%). Some other limitations discovered in the architecture are as follows:

1. *Architectural Redundancy and also Low Information Gain*: Both MobileNetV1 and MobileNetV2 belong to the same lightweight architectural family, relying heavily on depthwise separable convolutions and inverted residual bottleneck blocks. Because their feature extraction mechanisms are structurally similar, their feature representations overlap significantly. Fusing two similar models provides lower information gain compared to combining complementary, functionally specialized backbones.
2. *Premature Spatial Collapse*: Applying Global Average Pooling (GAP) before feature fusion flattens 7 x 7 2D spatial feature maps into 1D vectors. This discards 2D spatial coordinates prior to cross-feature interaction, preventing the model from performing explicit spatial query-key alignment across branches.
3. *Absence of Predictive Uncertainty Quantification*: The AFML network outputs single point-estimate

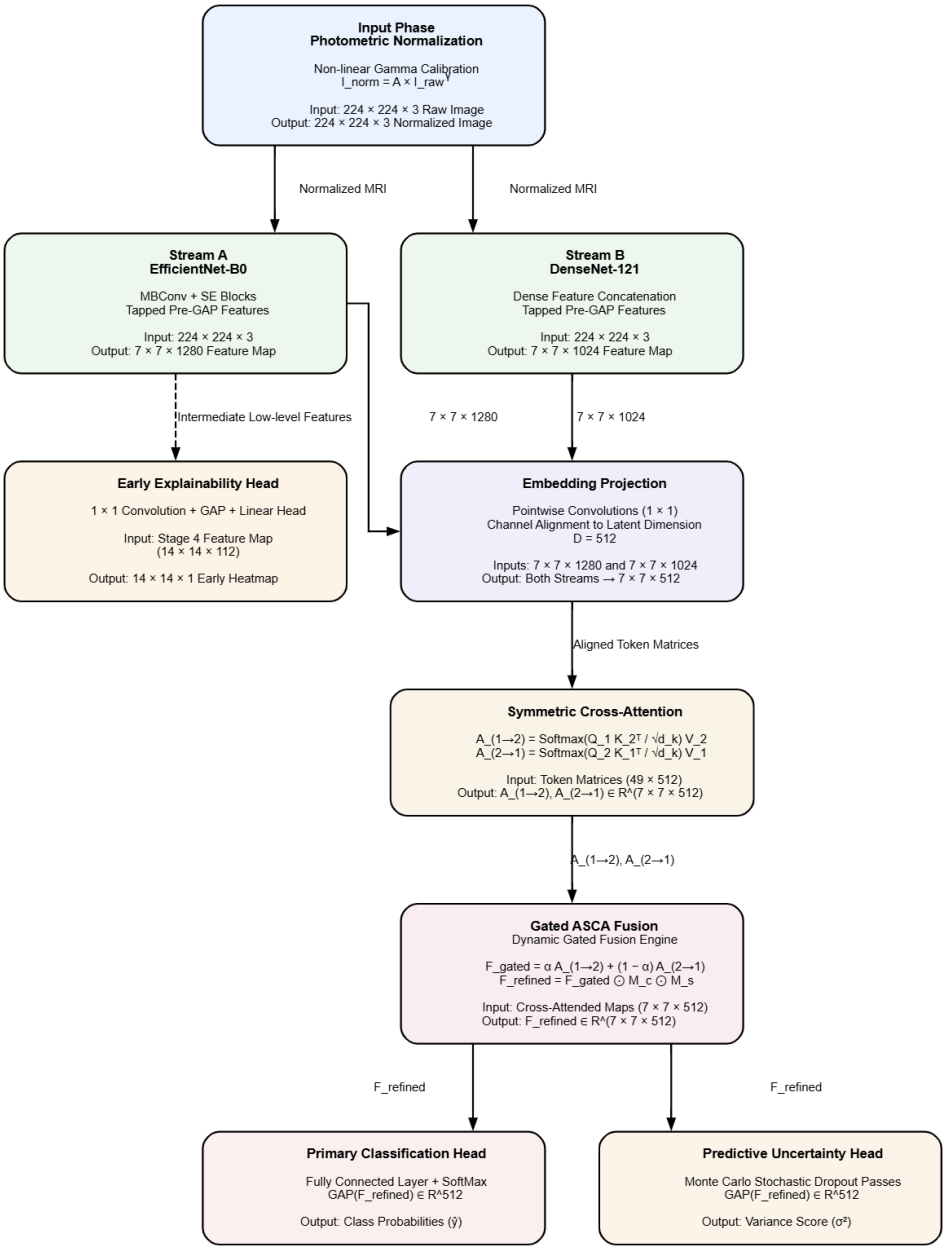


Figure 2: Proposed ED-XAGN architecture

class probabilities. It cannot measure predictive variance or clinical uncertainty, offering no mechanism to flag ambiguous scans for expert radiologist review.

Our proposed ED-XAF framework directly resolves these limitations by replacing the redundant MobileNet encoders with specialized EfficientNet-B0 and DenseNet-121 backbones, eliminating the LSTM in favor of 2D symmetric bidirectional cross-attention, and integrating an early single-stream parallel Grad-CAM explainability module attached directly to EfficientNet-B0.

3. Methodology

The proposed architecture, although sharing the same base as the AFML model, incorporates several key improvements which enables it to address the limitations of the baseline whilst being ultra-compact and keeping compute to a minimum. The modifications (in brief) are as follows:

1. *Backbone Replacement for Functional Specialization*: We replace MobileNetV1 and MobileNetV2 with two complementary, structurally distinct CNN backbones:
   (a) ***EfficientNet-B0 (Local Spatial and Texture)*** (13): Utilizes compound scaling (balancing depth,

width, and resolution) with mobile inverted bottleneck (MBConv) blocks and built-in Squeeze-and-Excitation (SE) channel attention (21). It extracts local boundary geometry, fine tumor edge transitions, micro-calcifications, and surface textures.

(b) ***DenseNet-121 (Multi-Scale Semantic)*** (14): Utilizes direct shortcut connections where every layer receives concatenated feature maps from all preceding layers. It preserves low-level, mid-level, and high-level feature maps simultaneously, capturing rich multi-scale semantic abstractions and long-range anatomical context.

2. *LSTM Removal*: We eliminate the LSTM module entirely. Instead of treating static spatial vectors as dynamic time series, feature interaction is handled by an explicit, transparent Bidirectional Cross-Attention Engine that operates directly on 2D spatial feature maps.
3. *Preservation of 2D Spatial Layout*: Feature maps are tapped prior to Global Average Pooling at their spatial resolution (7×7), maintaining 2D spatial coordinates throughout cross-attention querying and gated fusion.
4. *Single-Stream Early Parallel Grad-CAM Mechanism*: We eliminate the late-stage Grad-CAM overhead and replace it entirely with a single-stream early parallel Grad-CAM head attached directly to an intermediate stage of EfficientNet-B0 (14×14 resolution). This generates a high-resolution early Grad-CAM heatmap focused on local structural edge boundaries in parallel with the main cross-attention pipeline.
5. *Clinical Decision Support and Predictive Uncertainty*: The output stage incorporates a dual-head multi-layer perceptron (MLP) containing a primary classification head alongside an Uncertainty Estimation Head. Scans with high predictive variance are flagged automatically for escalation.

3.1. Dataset

For our research we utilized the publicly available Brain-Tumor-MRI-Dataset (15) on Huggingface. This dataset is composed of 7,200 images from various sources such as Br35H, figshare and SARTAJ. The sample distribution is given in more detail in 1, given below.

Table 1: Dataset distribution

Class	Number of images
No tumor	1,800
Glioma	1,800
Meningioma	1,800
Pituitary	1,800
Total	7,200

For the training set, a total of 5,600 images were included (1,400 from each category). Similarly for the testing set, 1,600 images were included (400 from each category). Care was taken to make sure no category was underrepresented in the final splits, so as to not skew the results.

3.2. Dataset Preprocessing and Augmentation

The dataset included images which were either too noisy for use in the model training or whose sizes varied too widely to be compatible with the CNN backbone of our architecture. To address these issues, as well as to diversify the dataset, we applied certain preprocessing and augmentation methods which are listed below:

1. *Rescale*: Each image in our dataset was rescaled by dividing the pixel values by 255. This normalization ensures that the images are within a standard dimension range of [0,1], preparing them for effective processing by the CNN.
2. *Rotations*: We introduced a 90-degree rotation for each image in the dataset. This augmentation technique generates additional variations of the original images. By doing so, the model becomes more adept at recognizing objects from various orientations, enhancing their robustness.
3. *Vertical and Horizontal Flipping*: Both vertical and horizontal flips were applied to the images. A horizontal flip creates mirror- image versions of each original image, which aids the model in recognizing objects from both left-to-right and right-to-left perspectives. Similarly, a vertical flip produces top-to-bottom mirrored versions of each image. This augmentation helps the model to learn object recognition from vertical reversals as well.

Using the above image augmentation techniques like rescaling, rotating, and flipping helps to increase the diversity of the training data, which is crucial for improving the robustness of a ML model. These techniques prevent overfitting by ensuring the model is trained on varied representations of the same images, thereby enhancing its ability to generalize to new, unseen data.

3.3. Proposed Architecture

At its core, the framework first standardizes contrast shifts across raw MRI scans using dynamic gamma calibration to mitigate spatial feature degradation and interpretability bottlenecks. The normalized input is simultaneously routed into two parallel, functionally specialized feature extraction streams- an EfficientNet-B0 backbone dedicated to isolating high-frequency, local spatial boundaries and related micro-textures, and a DenseNet-121 backbone designed to capture multi-scale semantic abstractions amongst other broad anatomical context.

To preserve spatial geometry and avoid premature information collapse, feature maps are tapped prior to Global Average Pooling (GAP)and projected into a shared latent space. Within this space, a symmetric bidirectional cross-attention module enables the streams to mutually query each other, aligning localized edge details with global context. These cross-attended feature maps are then integrated via

an Adaptive Spatial-Channel Gated Fusion (ASCA) module, which dynamically balances stream contributions and recalibrates spatial and channel relevance on an image-by-image basis. Finally, ED-XAGN incorporates a single-stream early parallel GradCAM head attached directly to an intermediate stage of the EfficientNet-B0 backbone to generate high-resolution diagnostic heatmaps, alongside a dual-head output stage that couples primary class predictions with a dropout driven predictive uncertainty head to quantify decision confidence and flag ambiguous scans for expert review.

3.3.1. Dynamic Gamma Normalization

The input MRI is resized to 224 × 224 pixels and represented as a three-channel image. Pixel intensities are rescaled to [0, 1] before gamma correction:

$$I_{\text{norm}} = A\, I_{\text{raw}}^{\gamma} \tag{1}$$

with $A = 1.0$ and $\gamma = 1.2$. Here I_{raw} is the rescaled input and I_{norm} is the gamma-corrected image. The transformation expands contrast around low-intensity tumor boundaries and peritumoral regions while preserving anatomical detail. Input and output sizes are both 224 × 224 × 3, and the normalized image is sent to both backbone streams.

3.3.2. Parallel Specialized Feature Backbone

The normalized tensor $I_{\text{norm}} \in \mathbb{R}^{224\times224\times3}$ is passed simultaneously to EfficientNet-B0 and DenseNet-121. In both streams the feature maps are tapped before GAP so that spatial layouts are preserved:

$$F_{\text{Eff}} \in \mathbb{R}^{7\times7\times1280} \tag{2}$$

$$F_{\text{Dense}} \in \mathbb{R}^{7\times7\times1024} \tag{3}$$

The EfficientNet-B0 stream uses MBConv and SE blocks and emphasizes local edge transitions, texture variations, and boundary geometry. The DenseNet-121 stream uses dense feature concatenation, reusing feature maps from preceding stages to preserve multi-level patterns and broader context.

3.3.3. Early GradCAM Explainability Head

The explainability pathway is connected to an intermediate EfficientNet-B0 stage (Stage 4). The tapped tensor has 14 × 14 resolution and 112 channels:

$$F_{\text{stage4}} \in R^{14\times14\times112} \tag{4}$$

A 1 × 1 convolution with batch normalization and ReLU reduces the channels from 112 to 64:

$$F_{\text{early}} = \text{ReLU}\big(\text{BN}\big(\text{Conv}_{1\times1}(F_{\text{stage4}})\big)\big) \in \mathbb{R}^{14\times14\times64} \tag{5}$$

GAP converts the intermediate features into a 64-dimensional vector, which a linear layer maps to auxiliary class scores:

$$\hat{y}_{\text{aux}} = \text{Softmax}\big(W_{\text{aux}}\ \text{GAP}(F_{\text{early}}) + b_{\text{aux}}\big) \tag{6}$$

For class c, the Grad-CAM weight of channel k is the average gradient of the auxiliary class score (23):

$$\alpha_k^c = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\frac{\partial y_{\text{aux}}^c}{\partial F_{\text{early},i,j}^k} \tag{7}$$

where $H = W = 14$. The class-specific heatmap is

$$L_{\text{GradCAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c F_{\text{early}}^k\right) \in \mathbb{R}^{14\times14} \tag{8}$$

The heatmap is bilinearly upsampled from 14 × 14 to 224 × 224 for visualization. This explanation pathway operates in parallel with the main classification path rather than requiring a final-stage explanation pass through the full cross-attention pipeline, and it focuses on local structural boundaries.

3.3.4. Linear Projection into Shared Embedding Space

The two feature maps have different channel counts (1280 and 1024), so both are projected to a common dimension $D = 512$ using pointwise 1 × 1 convolutions:

$$E_{\text{Eff}} = \text{Conv}_{1\times1}(F_{\text{Eff}}) \in \mathbb{R}^{7\times7\times512} \tag{9}$$

$$E_{\text{Dense}} = \text{Conv}_{1\times1}(F_{\text{Dense}}) \in \mathbb{R}^{7\times7\times512} \tag{10}$$

The spatial dimensions are flattened to form token matrices

$$X_{\text{Eff}}, X_{\text{Dense}} \in \mathbb{R}^{49\times512} \tag{11}$$

where $N = 7 \times 7 = 49$ spatial tokens. Attention outputs are reshaped back to spatial maps after cross-attention.

3.3.5. Symmetric Bidirectional Cross-Attention

Both streams query each other symmetrically (Fig. 3) using scaled dot-product attention (19). In the first direction, EfficientNet-B0 features query DenseNet-121 features:

$$Q_{\text{Eff}} = X_{\text{Eff}} W_{q1} \tag{12}$$

$$K_{\text{Dense}} = X_{\text{Dense}} W_{k2} \tag{13}$$

$$V_{\text{Dense}} = X_{\text{Dense}} W_{v2} \tag{14}$$

$$A_{1\to2} = \text{Softmax}\left(\frac{Q_{\text{Eff}} K_{\text{Dense}}^{\mathsf{T}}}{\sqrt{d_k}}\right) V_{\text{Dense}} \tag{15}$$

This direction uses local-detail queries from EfficientNet-B0 to find relevant information in the DenseNet-121 key/value representation. In the reverse direction, DenseNet-121 queries EfficientNet-B0:

$$Q_{\text{Dense}} = X_{\text{Dense}} W_{q2} \tag{16}$$

$$K_{\text{Eff}} = X_{\text{Eff}} W_{k1} \tag{17}$$

$$V_{\text{Eff}} = X_{\text{Eff}} W_{v1} \tag{18}$$

$$A_{2\to1} = \text{Softmax}\left(\frac{Q_{\text{Dense}} K_{\text{Eff}}^{\mathsf{T}}}{\sqrt{d_k}}\right) V_{\text{Eff}} \tag{19}$$

The reverse direction relates broader semantic information to fine spatial details from EfficientNet-B0. Eight attention heads are used with a projection scaling dimension $d_k = 64$. The outputs are reshaped into $7 \times 7 \times 512$ feature maps.

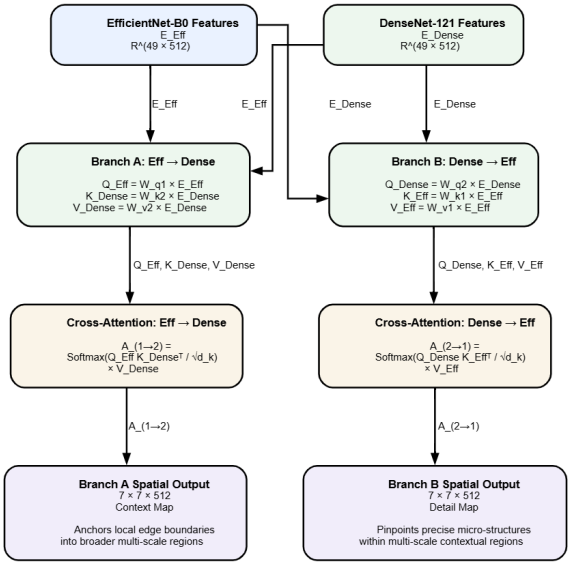


Figure 3: *Bi-directional Cross-Attention Module*

3.3.6. Adaptive Spatial-Channel Gated Fusion (ASCA)

The two cross-attended maps are combined using a learnable convex mixing parameter $\alpha \in [0, 1]$:

$$F_{\text{gated}} = \alpha A_{1\to2} + (1 - \alpha) A_{2\to1} \tag{20}$$

The gate lets the network adjust the relative contribution of each interaction direction on an image-by-image basis. The gated tensor is then refined by the ASCA module (Fig. 4), which applies channel and spatial attention in parallel.

Channel attention. GAP summarizes spatial context, and a two-layer bottleneck produces channel weights (21):

$$M_c(F) = \sigma\big(W_2\, \delta\big(W_1\ \text{GAP}(F)\big)\big) \tag{21}$$

where δ is ReLU, σ is the sigmoid function, W_1 and W_2 are learned transformations, and the reduction ratio is $r = 16$. The channel weights have dimensions $1 \times 1 \times 512$

Spatial attention. The tensor is pooled across channels by mean and maximum operations, concatenated, and passed through a 7×7 convolution and sigmoid (22):

$$M_s(F) = \sigma\Big(\text{Conv}_{7\times7}\left([\text{MeanPool}(F) \parallel \text{MaxPool}(F)]\right)\Big) \tag{22}$$

where $\parallel$ denotes channel concatenation. The spatial mask has dimensions $7 \times 7 \times 1$.

Composite refinement. The refined tensor is

$$F_{\text{refined}} = F_{\text{gated}} \odot M_c(F_{\text{gated}}) \odot M_s(F_{\text{gated}}) \tag{23}$$

where $\odot$ denotes element-wise multiplication with broadcasting. The result remains a $7 \times 7 \times 512$ feature map. Channel recalibration weights useful channels, while spatial recalibration highlights locations considered relevant by the learned module.

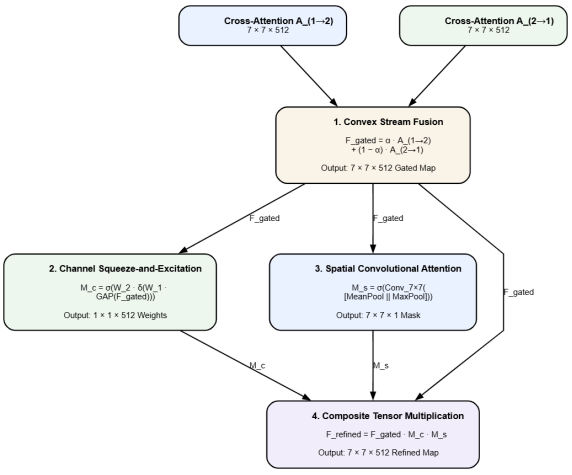


Figure 4: *ASCA Module*

3.3.7. Dual Classification and Uncertainty Head

GAP converts the refined feature map into a 512 dimensional vector:

$$z = \text{GAP}(F_{\text{refined}}) \in \mathbb{R}^{512} \tag{24}$$

The primary classification head applies a fully connected layer and Softmax to produce probabilities for the four classes:

$$\hat{p} = \text{Softmax}(W_c z + b_c) \tag{25}$$

The uncertainty head uses Monte Carlo dropout (24). With dropout enabled, the model performs $T = 20$ stochastic forward passes. Let $p_c^{(t)}$ be the probability of class c in pass t. The mean probability is

$$\bar{p}_c = \frac{1}{T} \sum_{t=1}^{T} p_c^{(t)} \tag{26}$$

and the per-class predictive variance is

$$u_c = \frac{1}{T} \sum_{t=1}^{T} \left(p_c^{(t)} - \bar{p}_c\right)^2 \tag{27}$$

A summary variance score is obtained by averaging across the $C = 4$ classes:

$$u = \frac{1}{C} \sum_{c=1}^{C} u_c \tag{28}$$

The dropout rate is $p = 0.3$, and scans whose uncertainty exceeds a threshold are proposed for manual radiologist review. The threshold must be chosen on validation data; no numerical value is specified here. The variance is an uncertainty indicator and should not be interpreted as a calibrated probability of error without additional validation.

3.4. Module-wise Tensor Dimensions

Table 3.4 summarizes the tensor shapes along the main pipeline and the auxiliary explainability branch.

Table 2: Tensor Dimensions in ED-XAGN

Stage	Shape
Normalized input I_{norm}	$224 \times 224 \times 3$
EfficientNet-B0 output F_{Eff}	$7 \times 7 \times 1280$
DenseNet-121 output F_{Dense}	$7 \times 7 \times 1024$
Stage-4 tap F_{stage4} (aux.)	$14 \times 14 \times 112$
Reduced F_{early} (aux.)	$14 \times 14 \times 64$
Projected $E_{\text{Eff}}, E_{\text{Dense}}$	$7 \times 7 \times 512$
Token matrices $X_{\text{Eff}}, X_{\text{Dense}}$	49×512
Cross-attended $A_{1\to2}, A_{2\to1}$	$7 \times 7 \times 512$
Refined F_{refined}	$7 \times 7 \times 512$
Pooled vector z	512
Class probabilities $\hat{p}$	4

3.5. Hyperparameter Configuration

The following settings are specified for implementation in PyTorch (28) (Table 3.5).

Table 3: Hyperparameter Configuration

Parameter	Configuration
Input image resolution	$224 \times 224 \times 3$ (bilinear)
Optimizer (26)	Adam
Initial learning rate	10^{-4}
Learning-rate schedule (27)	Cosine annealing
Batch size	32
Training duration	30 epochs
Attention heads	$H = 8$ ($d_k = 64$)
Shared latent dimension	$D = 512$
Auxiliary bottleneck channels	64
Monte Carlo dropout passes	$T = 20$
Dropout rate	$p = 0.3$
Gamma correction	$A = 1.0$, $\gamma = 1.2$
Loss function (25)	Composite focal loss

The composite focal-loss (25) objective combines the primary and auxiliary classification losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{primary}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} \tag{29}$$

where λ_{aux} is the auxiliary-loss weight.

4. Implementation and Evaluation

4.1. Evaluation Metrics

Accuracy, precision, recall, and F1-score are used. For a test set of N samples and C classes, overall accuracy is

$$\text{Accuracy} = \frac{\sum_{c=1}^{C} TP_c}{N} \tag{30}$$

where TP_c is the number of correctly classified samples of class c. For each class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \tag{31}$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{32}$$

$$F1_c = 2\,\frac{\text{Precision}_c\,\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{33}$$

where FP_c and FN_c denote false positives and false negatives for class c.

4.2. Results

The results of the training run are as follows:

5. Conclusion and Future Work

5.1. Conclusion

In this work, we presented ED-XAGN, a novel deep learning framework designed to overcome the spatial information collapse, feature redundancy, and opacity inherent in conventional computer-aided brain tumor diagnosis systems. By replacing uniform backbones with two functionally specialized feature extractors—EfficientNet-B0 for fine local boundaries and micro-textures, and DenseNet-121 for multi-scale semantic context—the model captures comprehensive pathological representations. Preserving 2D spatial coordinate geometry prior to global pooling allows the symmetric bidirectional cross-attention engine to mutually align localized spatial queries with broad anatomical structures, while the Adaptive Spatial-Channel Gated Fusion (ASCA) engine dynamically recalibrates spatial and channel relevance on a scan-by-scan basis. Crucially, ED-XAGN bridges the gap between predictive performance and clinical usability by coupling an early parallel Grad-CAM head for high-resolution edge attributions with a Monte Carlo dropout-based uncertainty head that quantifies decision confidence and automatically escalates ambiguous scans for expert radiologist review. Together, these innovations establish ED-XAGN as an accurate, interpretable, and clinically self-aware solution for automated neuro-oncological assessment.

5.2. Future Work

While ED-XAGN demonstrates strong predictive performance and clinical interpretability, several routes exist for future investigation:

1. *Cross-Institutional Validation*: Evaluating the architecture across multi-center, cross-institutional MRI datasets will validate its domain generalization capabilities against varying hospital scanner protocols and artifact distributions.
2. *3D Volumetric and Multi-Sequence Extension*: Extending the 2D slice-level cross-attention mechanism to 3D volumetric MRI representations, which would involve incorporating multi-sequence modalities such as T1-weighted, T2-weighted, and FLAIR scans—could provide richer 3D spatial context.
3. *Active Learning and Human-in-the-Loop Integration*: Integrating the model's predictive uncertainty scores into an active learning framework will allow high-variance predictions to be automatically flagged for

radiologist annotation, allowing continual fine-tuning in clinical environments.

4. *Model Optimization and Edge Deployment*: Applying structural pruning and knowledge distillation to compress the dual-backbone architecture will further reduce computational complexity, supporting low-latency deployment on edge devices.

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